Perspectives from a Workshop: Intelligent Assessment in the Age of Artificial Intelligence

Abstract

The advent of generative artificial intelligence (GenAI) is already impacting pedagogical strategies and assessment methodologies in higher education, particularly in the biological sciences, which have traditionally relied heavily on written assessments. GenAI's rapid and plausible text generation capabilities challenge traditional written assessments and prompt a shift toward more authentic assessment types. This article explores innovative applications of GenAI in biology education through case studies presented at a recent workshop. These case studies illustrate how GenAI has the potential to enhance academic activities, from developing learning resources to fostering student engagement through active learning strategies. The discussion highlights a shift from product-oriented assessments to process-oriented approaches that prioritize continuous interaction, iteration, and reflection among learners. Despite GenAI's reliance on preexisting data, raising concerns about originality and contextual accuracy, and its limitations in tasks requiring high creativity and deep understanding, it has the potential to enhance educational practices when applied with awareness of its constraints. The article concludes with a balanced analysis of the transformative impact and inherent challenges of integrating GenAI into biology education, advocating for thoughtful implementation to ensure it augments rather than replaces traditional teaching methods. Generative artificial intelligence (GenAI) is transforming higher education by enabling rapid learning resource development, enhancing student engagement, and supporting authentic assessment. Our workshop-based case studies highlight GenAI's ability to foster interactive, process-oriented learning in biosciences while addressing challenges with creativity and originality. From creating tailored quizzes to promoting active learning and ethical AI use, these strategies empower educators to integrate AI responsibly, ensuring it enriches teaching and learning in bioscience education while maintaining academic integrity.